

Lec 4. Series Solution

Variation of constants/parameters



(9) Power series near x_0

$$y(x) = \sum_{n=0}^{\infty} a_n (x-x_0)^n$$

Convergence radius.

$$|x-x_0| < \mu^{-1}, \quad \mu = \limsup |a_n|^{\frac{1}{n}} < +\infty$$

$y' = \sum_{n=0}^{\infty} n a_n (x-x_0)^{n-1}$ with the same convergence radius

(10) Recall.

$$P(x) y'' + Q(x) y' + R(x) y = 0 \quad (1)$$

If $P(x_0) \neq 0$, x_0 is an ordinary point.

$P(x_0) = 0$, x_0 is a singular point

consider solving (1) near x_0 with.

$$y(x) = \sum_{n=0}^{\infty} a_n (x-x_0)^n$$

E.g. Airy's equation

$$y'' - xy = 0 \quad \text{near } x_0 = 0$$

$$y = \sum_{n \geq 0} a_n x^n$$

$$y' = \sum_{n \geq 0} a_n n(n-1) x^{n-2} = \sum_{n \geq 2} a_n n(n-1) x^{n-2}$$

$$\xrightarrow{n \rightarrow n+2} \sum_{n \geq 0} a_{n+2} (n+2)(n+1) x^n$$

$$x y = \sum_{n \geq 0} a_n x^{n+1} = a_0 x + \sum_{n \geq 0} a_{n+1} x^{n+1}$$

Match power of

$$x^0: \quad a_2 \cdot 2 \cdot 1 - a_0 = 0, \quad a_2 = \frac{a_0}{2}$$

$$x^n, n \geq 1: \quad a_{n+2} (n+2)(n+1) - a_{n-1} = 0$$

$$\Rightarrow \quad a_{n+2} = \frac{a_{n-1}}{(n+2)(n+1)}$$

$$\Rightarrow \quad a_{3k+i} = \frac{a_{3(k-1)+i}}{(3k+i)(3k+i-1)}$$

$$i = 0, 1, 2$$

$$k \geq 1$$

$$\boxed{i=0}$$

$$\Rightarrow$$

$$a_{3n} = \prod_{1 \leq k \leq n} \frac{1}{3k(3k-1)} \cdot a_0, \quad n \geq 1$$

$$a_{3n+1} = \prod_{1 \leq k \leq n} \frac{1}{(3k+1)3k} \cdot a_1$$

$$a_{3n+2} = \prod_{1 \leq k \leq n} \frac{1}{(3k+2)(3k+1)} \cdot \frac{a_0}{2}$$

$$(a_2 = \frac{a_0}{2})$$

$$\begin{aligned}
 \Rightarrow f(t) &= \sum_{n \geq 0} \left[a_0 \prod_{1 \leq k \leq n} \frac{1}{3k(3k+1)} \right] t^{3k} \\
 &+ a_1 \prod_{1 \leq k \leq n} \frac{1}{(3k+1)3k} \cdot t^{3k+1} \\
 &+ \frac{a_0}{2} \cdot \prod_{1 \leq k \leq n} \frac{1}{(3k+2)(3k+1)} t^{3k+2} \\
 \prod_{1 \leq k \leq 0} 1 &= 1 \quad \checkmark
 \end{aligned}$$

Convergence Radius. $i = 0, 1, 2$

$$\prod_{1 \leq k \leq n} \frac{1}{3k+1} \leq 1$$

$$\text{so } \mu = \lim_{n \rightarrow \infty} \left[a_{3n+i} \right]^{\frac{1}{n}} \leq 1. \quad i = 0, 1, 2.$$

convergence radius at least for $|x| < \mu$
 $\mu \geq 1$.

Consider: $y'' - xy' = 0$, near $x=1$.

Assume.

$$y = \sum_{n=0}^{\infty} a_n (x-1)^n.$$

$$\begin{aligned} y'' &= \sum_{n=2}^{\infty} a_n n(n-1) (x-1)^{n-2} \\ &= \sum_{n=2}^{\infty} a_{n+2} (n+2)(n+1) (x-1)^n \end{aligned}$$

$$\begin{aligned} xy' &= (x-1+1) \sum_{n=0}^{\infty} a_n (x-1)^n \\ &= \sum_{n=0}^{\infty} [a_n (x-1)^{n+1} + a_n (x-1)^n] \\ &= a_0 + \sum_{n=1}^{\infty} [a_{n-1} + a_n] (x-1)^n \end{aligned}$$

Match power: [Recursive Relation].

$$(x-1)^0: a_2 \cdot 2 \cdot 1 - a_0 = 0$$

$$n \geq 1 \quad (x-1)^n: a_{n+2} (n+2)(n+1) - a_{n-1} - a_n = 0$$

$$\Rightarrow a_2 = \frac{a_0}{2 \cdot 1}$$

$$a_{n+2} = \frac{a_{n-1}}{(n+1)(n+2)} + \frac{a_n}{(n+2)(n+1)}$$

$$a_3 = \frac{a_0 + a_1}{2 \cdot 3}$$

$$a_4 = \frac{a_1 + a_2}{3 \cdot 4} = \frac{a_0}{2 \cdot 3 \cdot 4} + \frac{a_1}{3 \cdot 4}$$

The recursive relation is complicated but
and,

$$y = C_0 \left[1 + \frac{(x-1)^2}{2 \cdot 1} + \frac{(x-1)^3}{2 \cdot 3} + \frac{(x-1)^4}{2 \cdot 3 \cdot 4} + \dots \right] \\ + a_1 \left[(x-1) + \frac{(x-1)^3}{2 \cdot 3} + \frac{(x-1)^4}{3 \cdot 4} + \dots \right]$$

The form of an ansatz explicit.

set $C_0 = 1$, $a_1 = 1$. get $y_1(x)$

set $C_0 = 1$, $a_1 = 1$ get $y_2(x)$.

• Convergence guaranteed by Thm below.

• General near regular point, i.e. $\phi'(x_0) \neq 0$

$$\textcircled{1} \quad p(x) y'' + Q(x) y' + R(x) y = 0$$

or

$$\textcircled{1}' \quad y'' + p(x) y' + q(x) y = 0, \quad p = \frac{Q}{p}, \quad q = \frac{R}{p}$$

• Theorem (Thm 5.3.1 [BDM]) .

If $p = \frac{Q}{p}$, $q = \frac{R}{p}$ is analytic near x_0 with convergence radius at least R_0

• Then the general solution of $\textcircled{1}'$ ($\Leftrightarrow \textcircled{1}$) is

$$\textcircled{2} \quad y(x) = a_0 y_1(x) + a_1 y_2(x) = \sum_{n=0}^{\infty} a_n (x-x_0)^n$$

a_0, a_1 are arbitrary, and y_1, y_2 are two power series solution with radii of convergence at least R_0 . i.e.

$\textcircled{2}$ Converge for any x with $|x-x_0| < R_0$.

[No proof.]

• Application: Give example of p, Q, R which are analytic.

(II). Variation of parameters / constants.
to solve nonhomogeneous problem.

- Given 2 solutions $y_1(t)$ to

$$y'' + py' + qy = 0 \quad [\text{Homo.}] \quad (1)'$$

General Solution:

$$y(t) = C_1 y_1(t) + C_2 y_2(t). \quad (3)$$

Goal: Find a solution to

$$y'' + py' + qy = f(t) \quad (4)$$

with general $f(t)$.

Ex: we know

$$y'' + 5y' + 6y = 0, \quad y_1 = e^{-2t}, y_2 = e^{-3t}$$

Now to solve

$$y'' + 5y' + 6y = e^t.$$

Thm [Thm 3.5-1 [BPM]] If ϕ, ψ solve (2), then
 $\phi - \psi$ solves (1)', and $\phi - \psi = C_1 y_1(t) + C_2 y_2(t)$.

proof:

$$\phi'' + p\phi + q\phi = f$$

$$\psi'' + p\psi + q\psi = f$$

$$\Rightarrow (\phi - \psi)'' + p(\phi - \psi)' + q(\phi - \psi) = 0$$

Since the general solution of (1)' is given by
 $C_1 y_1(t) + C_2 y_2(t)$

and $\phi - \psi$ solves (1)', we get
 $\phi - \psi = C_1 y_1(t) + C_2 y_2(t)$.

As a result, once we find 1 solution $\gamma(t)$ to (4),
general solution to (2) is given by

$$y(t) = C_1 y_1 + C_2 y_2 + \gamma(t).$$

General solu to Nonhomo

= general solu to homo +

1 solution to nonhomo.

• Variation of parameters. Consider solve ④ by

$$y(t) = u_1(t) y_1(t) + u_2(t) y_2(t).$$

Q: Why introduce 2 new unknowns?

While we only have 1 equation?!

(24)

A: More degree of freedom, simpler form for $u_i(t)$.

$$y' = \underline{u_1'} y_1 + u_1 y_1' + \underline{u_2'} y_2 + u_2 y_2'$$

• Impose: $u_1' y_1 + u_2' y_2 = 0$ (5).a

• Then $y = u_1 y_1 + u_2 y_2$
 $y' = u_1 y_1' + u_2 y_2'$

$$y'' = u_1' y_1' + u_1 y_1'' + u_2' y_2' + u_2 y_2''$$

$$g(t) = y'' + p y' + q y$$

$$= u_1 [y_1'' + p y_1' + q y_1] + u_2 [y_2'' + p y_2' + q y_2] \\ + u_1' y_1' + u_2' y_2' \quad [\text{key: no } u_1'', u_2'' \text{ terms}].$$

$\Rightarrow g(t) = u_1' y_1' + u_2' y_2'$ (5).b

⑤.a. ⑤.b linear eqn on $u_1', u_2' \Leftrightarrow$

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ g(t) \end{pmatrix} \Leftrightarrow M \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ g(t) \end{pmatrix}$$

y_1, y_2 2 different solutions $\Rightarrow \det M \neq 0$
(to be proved later)

Thus take
$$\begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = M^{-1} \begin{pmatrix} 0 \\ f(t) \end{pmatrix}$$

Further integrate $\Rightarrow u_1(t), u_2(t)$.

E.g. $y'' + 5y' + 6y = e^t$ 

[optional]. Not required.

Time permitted: Euler eqn; special case $p(x) = 0$

Consider:

$$L[y] = x^2 y'' + \alpha x y' + \beta y = 0. \quad (6)$$

Try solution $y = x^r$

$\Rightarrow y$ solves (6) $\Leftrightarrow$

$$r(r-1) + \alpha r + \beta = 0$$

In general for r_1, r_2

$y_1 = x^{r_1}$, $y_2 = x^{r_2}$ are 2 solns to (6)